Physics
Standard and higher level
Paper 2
B.3 Gas laws

12 October 2026

W4ss Academy Paper 2 practice 1 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–P2–01

Student name

50 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for this examination paper is **[30 marks]**.

Answer **all** questions. Answers must be written within the answer boxes provided.

1. A container of volume 0.36 m^3 holds 2.0 mol of oxygen gas at a temperature of 260 K. Assume that the oxygen behaves as an ideal gas.

Molar mass of oxygen = 32 g mol^{-1}

(a) Calculate the pressure of the gas in the container. [1]

(b) Determine the mass of the gas in the container. [1]

(c) Calculate the average translational speed of the gas particles. [1]

(This question continues on the following page)

(Question 1 continued)

(d) The temperature of the gas in the container is increased.

Explain, using the kinetic theory, how this change leads to a change in pressure in the container. [4]

2. A container holds a sample of air at a pressure of 1.8×10^5 Pa. The density of the air in the container is 2.0 kg m^{-3}. Assume that the air behaves as an ideal gas.

(a) Calculate the average translational speed of air molecules. [2]

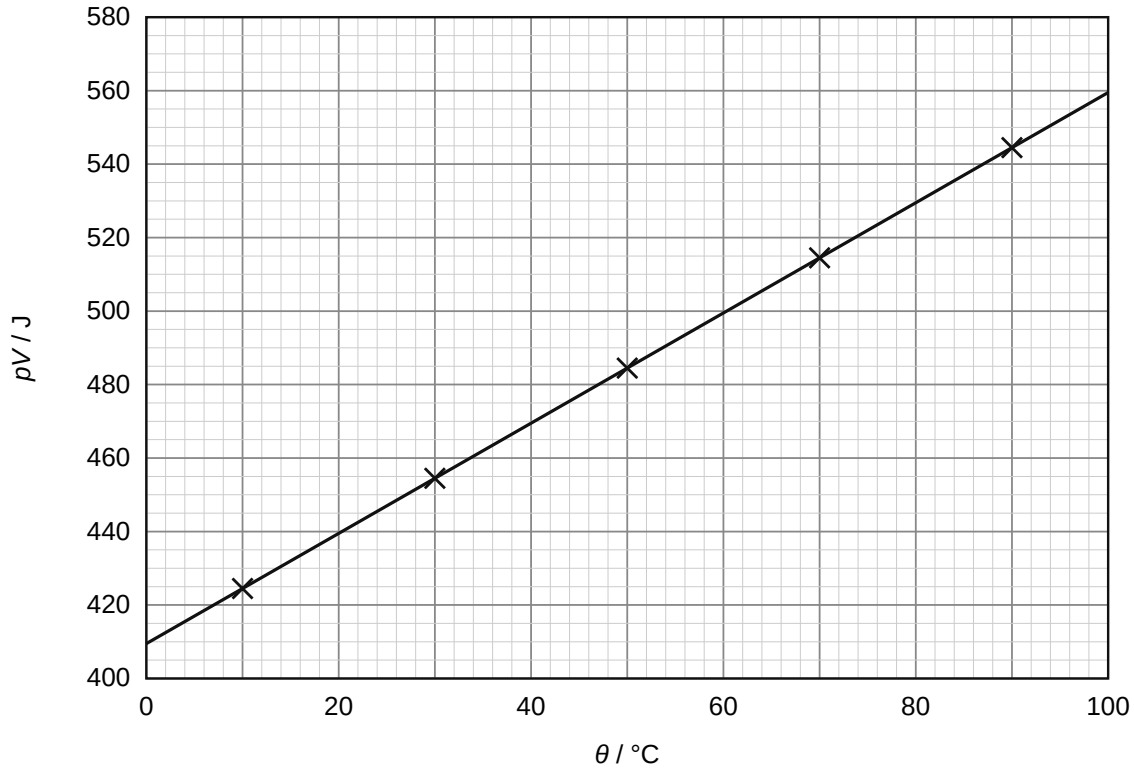
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(Question 2 continued)

(b) The air is a mixture of nitrogen, oxygen and other gases. Explain why the component gases of air in the container have different average translational speeds. [3]

(c) The temperature of the sample is increased without a change in pressure. Outline the effect it has on the density of the gas. [2]

3. A fixed mass of a gas is kept in a container. The graph shows how the product pV of the pressure p and the volume V of the gas varies with its temperature θ in °C.



(a) State the unit for pV in fundamental SI units. [1]

(b) Determine, using the graph, whether the gas acts as an ideal gas. [3]

(This question continues on the following page)

(Question 3 continued)

The mass of one molecule of the gas is 4.7×10^{-26} kg.

(c) Calculate, in g, the mass of the gas. [3]

4. A solar heating panel on the roof of a house heats the water in a storage tank. The tank contains 250 kg of water at an initial temperature of 15 °C.

The intensity of the solar radiation incident on the panel is 680 W m^{-2}. The panel transfers 30 % of the incident solar energy to the water in the tank.

Specific heat capacity of water = 4200 J kg^{-1} K^{-1}

(a) (i) Determine the minimum area of the solar heating panel required to increase the temperature of all the water in the tank to 30°C during a time of 1.0 hour. [3]

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(Question 4 continued)

The roof is covered with tiles that receive the same solar radiation. The albedo of the tiles is 0.20 and their emissivity is 0.97. Assume that the tiles are in thermal equilibrium and that they lose energy only by radiation from their upper surface.

(ii) Estimate, in °C, the temperature of the roof tiles. [3]

The tank is not completely full: the space above the water contains air. A small opening at the top of the tank keeps this air at atmospheric pressure, and the level of the water in the tank remains constant.

(b) (i) State **one** way in which a real gas differs from an ideal gas. [1]

(ii) The water is heated. Explain why the quantity of air in the storage tank decreases. [2]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.3 Gas laws. They remain the property of the International Baccalaureate Organization and must not be published or sold.

The introductions of these questions are missing from the Questionbank. W4ss Academy wrote them using only the data of the official markscheme, so that the markscheme still applies.

Paper 2 — question 1	EXE.2.SL.TZ0.10 — introduction written by W4ss Academy from the markscheme
Paper 2 — question 2	EXE.2.SL.TZ0.8 — introduction written by W4ss Academy from the markscheme
Paper 2 — question 3	23M.2.SL.TZ2.2 — introduction written by W4ss Academy from the markscheme (graph drawn by W4ss Academy)
Paper 2 — question 4	22N.2.SL.TZ0.2 — introduction written by W4ss Academy from the markscheme